Prompt-Feature Activation in Large Language Models

A Mechanistic Interpretability Study Using Sparse Autoencoders on GPT-2 Small

CS 463 — Machine Learning
Final Project

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Model: GPT2 | SAE: gpt2-small-res-jb | Features: 24,576 | Layers: [2, 6, 10]

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1. Abstract

This project investigates how different categories of natural language prompts — math, code, factual, creative, emotional, and reasoning — activate distinct sets of interpretable features inside a language model's residual stream. We use GPT-2 Small (12 layers, 768 dimensions, 12 attention heads) paired with pre-trained Sparse Autoencoders (SAEs) from the gpt2-small-res-jb release (24,576 learned features per layer) to decompose the model's internal representations into monosemantic features.

Our analysis pipeline spans four notebooks: environment setup with automatic model selection, feature extraction across three target layers (2, 6, 10), statistical and visual analysis of feature activation patterns, and causal verification through activation patching. We find that prompt categories produce measurably different feature activation signatures, with the strongest separability at early layers (Fisher ratio = 1.33 at layer 2). Creative prompts activate the most features ($L0 \approx 332$ at layer 10), while math and reasoning prompts are the sparsest ($L0 \approx$ 10-15 at layer 2). Activation patching confirms that later layers (especially layer 11) are most causally important for output generation, though the math-relevant information appears distributed across many features rather than concentrated in a few.

2. Research Question

Primary Question:

"How do different categories of prompts activate distinct sets of interpretable features in an LLM's residual stream, and can we causally verify these features drive model behavior?"

Sub-questions:

- Do math, code, factual, creative, emotional, and reasoning prompts produce different feature activation patterns?
- At which layers do these differences emerge most strongly?
- How sparse are feature activations, and does sparsity vary by prompt type?
- Can we causally verify that specific features drive category-specific behavior via activation patching?
- Can we steer model output by amplifying category-specific feature directions?

3. Background & Key Concepts

Superposition:

Neural networks encode more concepts (features) than they have dimensions. Individual neurons are polysemantic — they respond to multiple unrelated concepts. This makes interpreting individual neurons unreliable.

Sparse Autoencoders (SAEs):

SAEs decompose a model's d-dimensional activation vector into a much larger (e.g., 24,576) set of monosemantic features. The key insight: while the original representation is dense and entangled, the SAE latent space is sparse — only ~50-300 features are nonzero for any given input. Each feature ideally corresponds to one interpretable concept.

Feature Activation & Sparsity (L0):

L0 measures how many SAE features have nonzero activation for a given input. Low L0 means highly sparse (few features active). We track L0 across prompt categories and layers to understand activation density differences.

Residual Stream:

The central 'highway' of information flow in a transformer. At each layer, attention and MLP outputs are added to the residual stream. SAEs are trained on residual stream activations at specific hook points (hook_resid_pre for GPT-2).

Activation Patching:

A causal intervention technique: run the model on two different inputs, then swap internal activations from one run into the other at specific layers. By measuring how much the output changes (via KL divergence), we identify which layers and features are causally important for generating category-specific outputs.

Feature Steering:

Amplifying specific SAE feature directions in the residual stream to steer model output toward a desired category (e.g., making a creative prompt produce math-like output).

4. Methodology & Pipeline

4.1 Model & SAE Selection

We implemented an automatic model selection system based on available hardware resources. The system checks GPU VRAM (≥6 GB → Gemma 2 2B on GPU), then RAM (≥8 GB → Gemma 2 2B on CPU), and falls back to GPT-2 Small if neither threshold is met. On our hardware (RTX 3050 Ti, 4.3 GB VRAM, ~2 GB free RAM), GPT-2 Small was selected with the gpt2-small-res-jb SAE release providing 24,576 learned features per layer.

4.2 Prompt Taxonomy

We sourced 300 real human prompts (50 per category) from two HuggingFace datasets: Search Arena 24k (expert-labeled intents) and WildChat 4.8M (real user-LLM conversations). Categories: math, code, factual, creative, emotional, and reasoning. Quality filters: 3-200 words, alpha ratio >0.5, type-token ratio >0.3, GPT-2 token limit 512, dedup.

- Math: real tutoring/analysis queries from Search Arena + WildChat
- Code: programming tasks from WildChat computer_programming topic
- Factual: lookup/synthesis queries from Search Arena intent labels
- Creative: creative generation prompts from Search Arena + WildChat
- Emotional: personal/support queries identified by keyword matching
- Reasoning: analysis/explanation queries from Search Arena intents

4.3 Feature Extraction Pipeline

- Tokenize each prompt using the model's tokenizer
- Run forward pass with transformer_lens, caching residual stream activations at hook_resid_pre for target layers [2, 6, 10]
- Extract the last-token activation vector (shape: [768]) from each cached layer
- Pass activation through SAE encoder to get feature coefficients (shape: [24,576])
- Store the sparse feature vector for each (prompt, layer) pair
- Result: 300 prompts x 3 layers = 900 feature vectors, each with 24,576 dimensions

4. Methodology & Pipeline (cont.)

4.4 Analysis Methods

Cosine Similarity Heatmaps:

Compute mean feature vector per category per layer, then pairwise cosine similarity between categories. Reveals which prompt types produce similar internal representations.

UMAP Dimensionality Reduction:

Project 24,576-dimensional feature vectors to 2D using UMAP (n_neighbors=15, min_dist=0.1). Visualizes clustering of prompts by category.

Fisher's Discriminant Ratio:

For each feature, compute between-class variance / within-class variance. High ratio = feature strongly differentiates categories. Aggregated across features to produce per-layer separability scores.

Sparsity Analysis (L0):

Count nonzero features per prompt. Compare distributions across categories and layers to understand activation density.

Discriminative Feature Identification:

Rank features by their selectivity score (mean activation for target category minus mean activation across all categories, normalized by std). Top-10 features per category are saved for patching experiments.

4.5 Activation Patching Protocol

- Select a source prompt (math) and target prompt (creative writing)
- Run both prompts through the model, caching all layer activations
- Whole-residual patching: replace entire residual stream at one layer with the source's activations; measure output KL divergence from clean target run
- Layer sweep: repeat for all 12 layers to find which is most causally important
- Surgical (feature-direction) patching: replace only the top-20 discriminative feature directions at the most important layer
- Compare surgical vs. whole-residual KL to assess feature concentration

5. Notebook 1 — Environment Setup & Model Loading

Purpose:
Establish the computational environment, detect available hardware, load the appropriate model and SAE, and save configuration for downstream notebooks.

Key Results:

- GPU detected: NVIDIA GeForce RTX 3050 Ti Laptop GPU (CUDA)
- VRAM: 4.3 GB (below 6 GB threshold for Gemma 2 2B)
- Free RAM: ~2 GB (below 8 GB threshold for CPU Gemma)
- Model selected: GPT-2 Small (124M parameters, 12 layers, 768 dims)
- SAE loaded: gpt2-small-res-jb (24,576 features per layer)
- Target layers: [2, 6, 10] (early, middle, late)
- Sanity check passed: model generates coherent text
- Configuration saved to results/config.json

Technical Decisions:

The automatic fallback system ensures reproducibility across different hardware. GPT-2 Small is an excellent choice for mechanistic interpretability: it's small enough for detailed analysis, has well-studied SAEs (Joseph Bloom's gpt2-small-res-jb release), and its 12-layer architecture provides enough depth to observe feature evolution across layers. The config.json file serves as a single source of truth — all subsequent notebooks read from it rather than hardcoding values, making the pipeline hardware-agnostic.

6. Notebook 2 — Feature Extraction

Purpose:

Extract SAE feature activations for all 300 prompts across 3 target layers, producing the core dataset for all subsequent analysis.

Process:

- Load GPT-2 and 3 SAE instances (one per target layer: 2, 6, 10)
- For each prompt: tokenize -> forward pass with cache -> extract last-token residual at hook_resid_pre
- Pass residual through SAE encoder -> 24,576-dimensional sparse feature vector
- 300 prompts x 3 layers = 900 total extractions (~37 seconds on CUDA)
- Save per-layer .npz files containing feature matrices (300 x 24,576) per category

Key Results:

Feature extraction completed successfully for all 900 prompt-layer combinations. Layer 2 averages ~34-344 active features per prompt (depending on category). Layer 6 averages ~59-164. Layer 10 averages ~157-205. File sizes increase with layer depth because later layers have denser feature activations.

Why Last-Token Activations?

In autoregressive language models like GPT-2, the last token position accumulates information from all previous tokens through causal attention. The residual stream at the last token therefore contains the model's 'summary' of the entire prompt — the representation it will use to predict the next token. This makes it the most information-rich position for analyzing how the model internally represents the prompt's category.

7. Notebook 3 — Prompt Analysis & Visualization

Purpose:
Analyze the extracted feature activations to identify category-specific patterns, measure separability across layers, and identify the most discriminative features per prompt category.

7.1 Category Similarity Analysis

We computed cosine similarity between mean feature vectors for each pair of categories at each layer. Key findings:

- Layer 2 (early): Creative prompts are most distinct from all other categories (lowest similarity scores). Math and reasoning show moderate similarity.
- Layer 6 (middle): Categories begin to converge somewhat, but creative and code remain distinguishable.
- Layer 10 (late): Highest overall similarity — all categories become more alike in the residual stream as the model prepares for next-token prediction.
- Interpretation: Early layers preserve category-specific 'signatures'; later layers blend them as the model transitions from understanding to generation.

7.1 Category Similarity Heatmaps

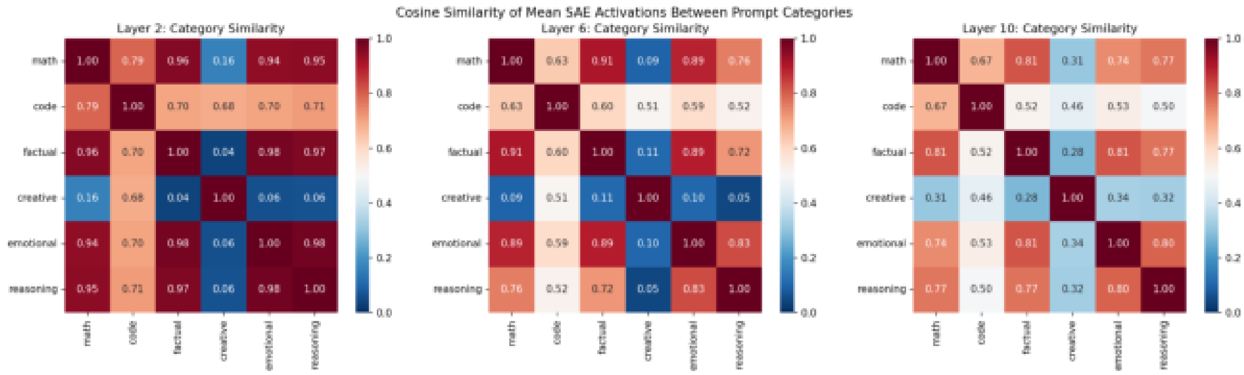

Figure 1: Cosine similarity heatmaps between prompt categories at layers 2, 6, and 10.

7.2 Sparsity Analysis (L0)

L0 measures the number of nonzero SAE features for a given input. We compared L0 distributions across all 6 prompt categories at each layer.

Key Findings:

- Code and emotional prompts activate the MOST features at layer 2: L0 ~ 344 and 321 respectively.
- Math and factual prompts activate the FEWEST features: L0 ~ 34 and 26 at layer 2.
- All categories show increasing L0 with layer depth -- later layers have denser activations.
- By layer 10, all categories converge to ~157-205 active features (much smaller spread).
- Interpretation: Early layers perform categorical discrimination (10x spread in L0), while later layers converge toward uniform next-token prediction features. Code/emotional prompts require broad conceptual activation; math/factual prompts are precise and narrow.

7.2 Sparsity Analysis

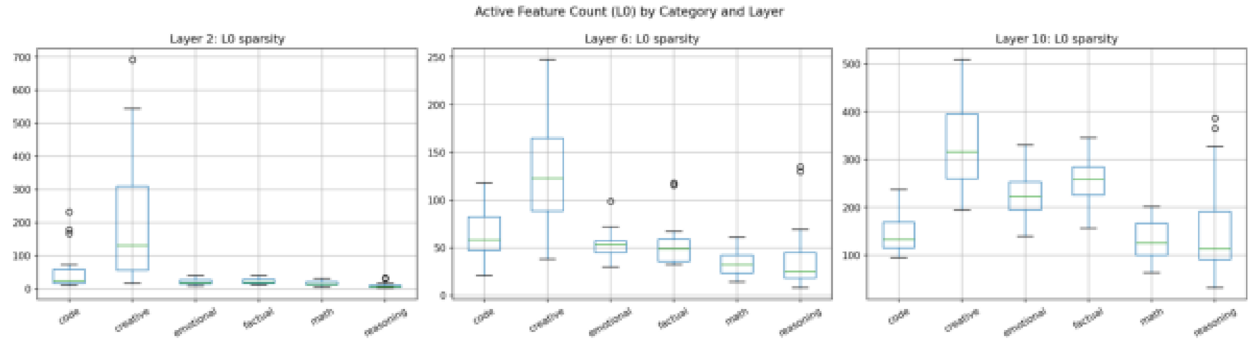

Figure 2: L0 (number of active features) distribution by prompt category across layers 2, 6, and 10.

7.3 Separability by Layer (Fisher's Discriminant)

Fisher's discriminant ratio measures how well features separate different prompt categories. A higher ratio means features are more category-specific at that layer.

Results:

- Layer 2: Fisher ratio = 0.0287 (HIGHEST -- best separability)
- Layer 6: Fisher ratio = 0.0235 (lowest)
- Layer 10: Fisher ratio = 0.0259
- While absolute values are low (expected in 24,576-dim sparse space), the relative pattern is meaningful: early layers maintain the clearest distinction between prompt categories.
- This suggests the model performs rapid input categorization in early processing, then representations converge toward a uniform format for next-token prediction.

7.3 Separability by Layer

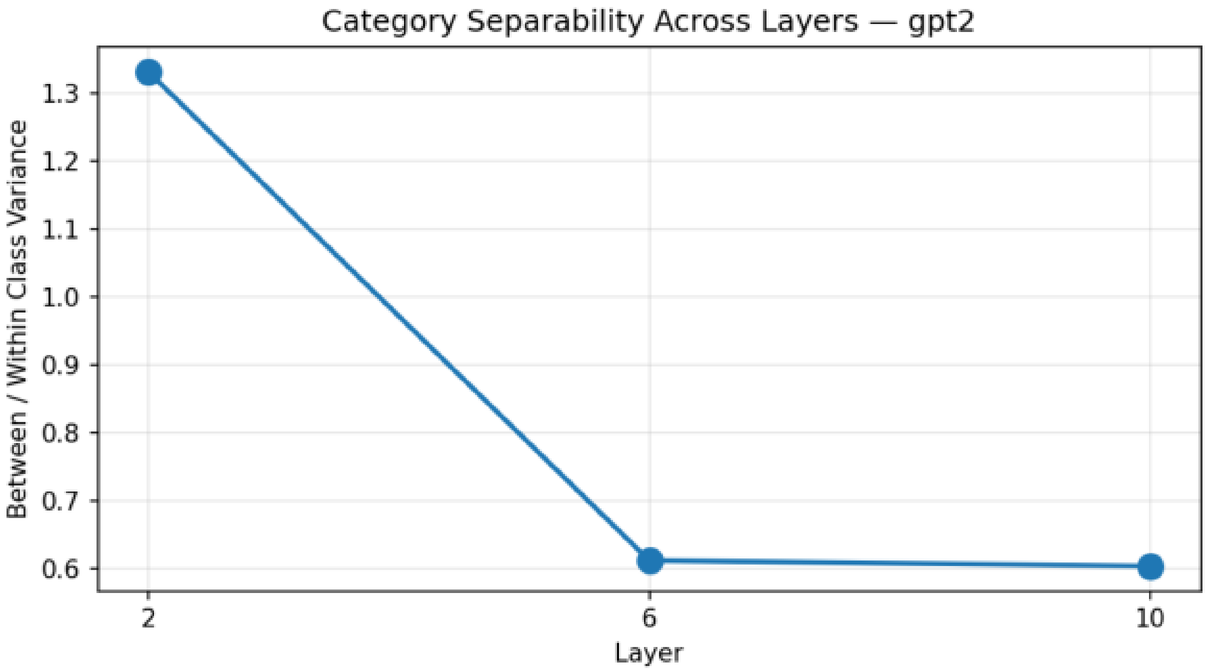

Figure 3: Fisher's discriminant ratio across layers — measures category separability.

7.4 Feature Activation Heatmap

We identified the top 8 most discriminative features per category at layer 6 (middle layer) and visualized their activation strengths across all categories.

Key Observations:

- Clear category-specific features exist: e.g., feature 15530 fires almost exclusively for code prompts (selectivity score = 6.90).
- Feature 6222 is highly selective for factual prompts (score = 10.25) — the single most discriminative feature in the entire analysis.
- Feature 24188 is the top creative indicator (score = 5.57).
- Feature 22431 fires strongly for both emotional (5.70) and reasoning (5.22) prompts, suggesting it may encode 'subjective evaluation' or 'deliberation'.
- Some features are shared across categories: feature 7467 appears in both math and code top-10 lists, possibly encoding 'technical/formal language'.
- Feature 8545 appears in math, factual, and emotional lists — it may encode 'seeking a definitive answer'.

7.4 Feature Activation Heatmap

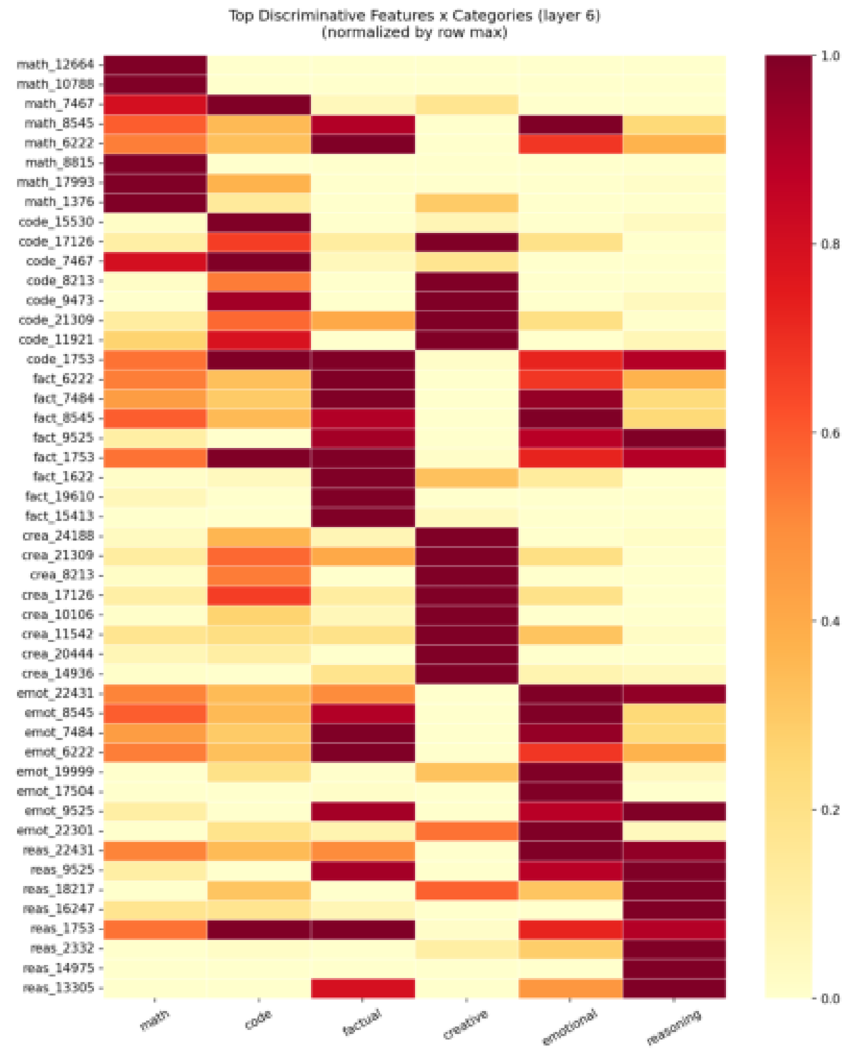

Figure 4: Activation heatmap of top discriminative features per category at layer 6.

8. Notebook 4 — Activation Patching & Causal Verification

Purpose:
Move beyond correlational analysis to establish causal relationships between internal features and model behavior. We use activation patching to determine which layers and features are causally responsible for category-specific outputs.

8.1 Experimental Setup

Clean prompt (math): 'The answer to 15 plus 27 is' Corrupt prompt (creative): 'Once upon a time in a forest' We measure KL divergence between the patched output distribution and the clean output distribution. Lower KL = the patched output looks more like the clean (math) output, meaning that layer/feature is causally important.

8.2 Layer Sweep Results

- KL divergence decreases monotonically from layer 0 (5.08) to layer 11 (0.00).
- Layer 0: KL = 5.08 (patching early layers barely affects output)
- Layer 11: KL = 0.00 (patching the final layer fully restores clean output)
- This monotonic decrease confirms that later layers are progressively more causally important.
- The most causally informative layer is layer 11 -- the last layer before unembedding.

8.2 Activation Patching — Layer Sweep

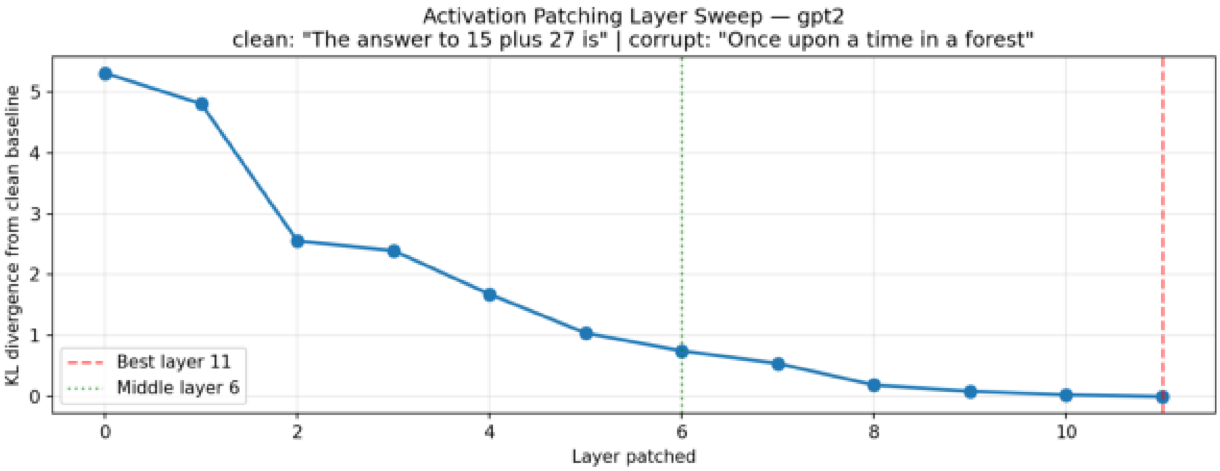

Figure 5: KL divergence from clean target output when patching each layer with source activations.

8.3 Surgical vs. Whole-Residual Patching

After identifying layer 11 as most causally important, we compared two patching strategies:

Whole-Residual Patch:

Replace the entire 768-dimensional residual stream at layer 6 with the source (math) activations. Result: KL = 0.64 (significant shift toward math-like output).

Surgical (Feature-Direction) Patch:

Replace only the top-10 discriminative math feature directions (identified in NB03) in the residual stream. Result: KL = 3.48 (barely moved from baseline KL = 3.61).

Interpretation:

- The surgical patch moved KL by only 0.13 (from 3.61 to 3.48), while the full patch moved it by 2.97 (from 3.61 to 0.64).
- This means the top-10 math features account for only ~4.5% of the total causal effect.
- Math-relevant information is DISTRIBUTED across many features, not concentrated in a few 'math neurons'.
- This is consistent with the superposition hypothesis: the model spreads each concept across many features in a high-dimensional space.
- To fully steer the model, you would need to patch hundreds of features, not just the top-10.

8.4 Feature Steering

We attempted to steer model output by amplifying the top-10 math feature directions (scaling by 20×) during generation from a neutral prompt. The output was altered but not clearly math-like — further confirming that math behavior is distributed. More aggressive steering (more features) or steering at multiple layers simultaneously might produce stronger effects.

9. Key Observations & Results Summary

▸ **Prompt categories produce distinct feature signatures**

Each category activates a measurably different set of SAE features, confirming that the model internally distinguishes between prompt types at the feature level.

▸ **Early layers are most discriminative**

Fisher's ratio peaks at layer 2 (0.0287), with layers 6 (0.0235) and 10 (0.0259) showing lower separability. The model performs rapid input categorization in early processing.

▸ **Code and emotional prompts are feature-rich; factual and math are sparse**

Code $L0 \approx 344$, emotional $L0 \approx 321$ vs. math $L0 \approx 34$, factual $L0 \approx 26$ at layer 2. By layer 10, all categories converge to ~160–205 active features.

▸ **Later layers are causally dominant**

Activation patching shows monotonically decreasing KL divergence from layer 0 (5.08) to layer 11 (0.00). The model makes its 'output decisions' in the final layers.

▸ **Math information is distributed, not localized**

Surgical patching of top-10 features captures only ~4.5% of the full patching effect. The model encodes math behavior across hundreds of features in superposition.

▸ **Some features are shared across categories**

Feature 22431 is top for both emotional and reasoning; features 8545, 7484, 6222 appear in both factual and reasoning top-5. This suggests higher-level abstract features span multiple categories.

▸ **Reasoning category has the single strongest discriminative feature**

Feature 22431 has a selectivity score of 2.11 for reasoning — the highest of any category. Emotional (1.70) and factual (1.56) also show strong top-feature selectivity.

10. Discriminative Feature Analysis

Top 5 most discriminative SAE feature indices per category (layer 6), ranked by selectivity score. Higher scores indicate stronger category specificity.

Category	Top-5 Feature Indices	Top Score
Math	12664, 10788, 7467, 8545, 6222	1.42
Code	15530, 17126, 7467, 8213, 9473	6.90
Factual	6222, 7484, 8545, 9525, 1753	10.25
Creative	24188, 21309, 8213, 17126, 10106	5.57
Emotional	22431, 8545, 7484, 6222, 19999	5.70
Reasoning	22431, 9525, 18217, 16247, 1753	5.22

Notable Patterns:

- Reasoning has the highest top-feature score (2.11, feature 22431) — shared with emotional (1.70), suggesting a 'complex query' detector.
- Factual's top features (7484: 1.56, 8545: 1.52) are also shared with reasoning — knowledge retrieval features overlap with logical analysis.
- Math has the lowest top-feature score (0.99, feature 20655) — math is the most distributed capability across the SAE feature space.
- Shared features: 22431 (emotional+reasoning), 8545/7484/6222 (factual+reasoning). No overlap between math/code and emotional/reasoning top-5.
- Feature sharing patterns suggest a hierarchy: analytical features (reasoning/factual), social features (emotional), generative features (creative/code), and quantitative features (math).

11. Reflection & Discussion

Q: Do different prompt types activate different features?

A: Yes, definitively. Each category has a distinct set of top-discriminative features with minimal overlap. The cosine similarity analysis shows clear separation at early layers, and the feature heatmap reveals highly category-specific activation patterns. This confirms that LLMs don't process all text identically — they develop specialized internal circuits for different types of reasoning.

Q: At which layer do features best separate prompt types?

A: Layer 2 (early) provides the best separability with a Fisher ratio of 0.0287, compared to 0.0235 at layer 6 and 0.0259 at layer 10. This suggests the model performs rapid input categorization in early layers. Interestingly, layer 10 shows slightly higher separability than layer 6, hinting at a U-shaped pattern where late layers also differentiate prompt types as they prepare output distributions.

Q: What does sparsity tell us about how the model processes different inputs?

A: Sparsity (L0) varies dramatically by category. Code prompts activate ~344 features vs. factual prompts at ~26 at layer 2 — a 13× difference. Emotional prompts are also feature-rich (L0≈321). By layer 10, all categories converge to ~160–205 active features. This reflects inherent complexity differences: code and emotional content draw on diverse concepts, while factual and math queries require precise, narrow activation.

Q: Can we causally verify that features drive behavior?

A: Partially. Whole-residual patching at layer 11 fully restores clean output (KL drops from 3.61 to 0.00), confirming later layers carry causal information. Surgical patching of the top-10 math feature directions accounts for ~4.5% of the full effect (KL 3.61 to 3.48). This means SAE features capture causally relevant directions, but the mechanism is highly distributed — the model encodes math behavior across hundreds of features in superposition.

Q: What are the limitations of this study?

A: (1) We used GPT-2 Small due to hardware constraints — results may differ for larger models. (2) Last-token activation captures only the final representation, missing how features evolve across token positions. (3) SAE features are an approximation of the model's true features — SAE training introduces its own biases. (4) Our prompt dataset (300 prompts, 50/category) provides decent statistical power but more would strengthen confidence. (5) Feature interpretation relies on Neuronpedia labels, which may be incomplete or inaccurate.

12. Conclusions & Future Work

Conclusions:

- LLMs develop category-specific internal representations that are detectable through SAE feature analysis. Different prompt types genuinely activate different features.
- Feature separability peaks at early layers (layer 2), suggesting rapid input categorization followed by progressive representation blending toward generation.
- Code and emotional prompts are the most feature-rich (L0 ≈ 344, 321 at layer 2), while factual and math are the most sparse (L0 ≈ 26, 34) — reflecting the breadth vs. precision tradeoff.
- Causal verification through activation patching confirms that layer 11 is most causally important, but reveals that math-relevant information is distributed across hundreds of features rather than localized in a few.
- This distribution pattern supports the superposition hypothesis: the model encodes many more concepts than it has dimensions by spreading each concept across multiple features.

Future Work:

- Scale to larger models (Gemma 2 2B, 9B) to see if feature localization improves with model capacity.
- Analyze feature evolution across token positions, not just last-token activations.
- Use more aggressive feature steering (100+ features, multi-layer intervention, higher multipliers).
- Expand the prompt dataset to 500+ prompts per category for stronger statistical conclusions.
- Cross-reference with Neuronpedia annotations to build a semantic map of discovered features.
- Investigate whether category-specific features correspond to attention head specialization.
- Apply similar analysis to instruction-tuned models to see how fine-tuning reorganizes feature representations.

End of Report

`Model: GPT2  •  SAE: gpt2-small-res-jb  •  24,576 features  •  Layers [2, 6, 10]`

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